

Final Exam Formulation
Math 5321

The final exam will consist of problems of the following types:

0. Statements of Definitions and Theorems from Review Part III

1. Classification of Isolated Singularities

Similar to problem 3 on Exam I In-Class or
Similar to problem 2 on Exam I Take-Home
Similar to problem 3 on 2013 August Prelim
Similar to problem 3 on 2013 May Prelim

2. Calculation of Residues at Poles

Similar to problem 4 on Exam I In-Class or
Similar to problem 3 on Exam I Take-Home
Similar to problem 3 on 2013 August Prelim
Similar to problem 3 on 2013 May Prelim

3. Construction of Laurent series expansions for rational functions

Similar to problem 1 on Exam I Take-Home
Similar to problem 7 on 2012 May Prelim
Similar to problem 1(b) on 2011 May Prelim
Similar to problem 5 on 2006 August Prelim

4. Computation of Integrals using the Residue Theorem

Similar to problem 4 on Exam I Take-Home
Similar to problem 1 on 1998 Final Exam
Similar to problem 6 on 2013 May Prelim
Similar to problem 2 on 2012 May Prelim

5. Applications of the Argument Principle or Rouché's Theorem

Similar to problem 4 on Exam II In-Class
Similar to problems 1-5 on Exam II Take-Home
Similar to problem 7 on 2013 August Prelim
Similar to problem 5 on 2012 May Prelim

6. Applications of Schwarz's Lemma

Similar to problem 5 on Exam II In-Class
Similar to problem 5 on Exam II Take-Home
Similar to problem 8 on 2013 May Prelim

6. Applications of Schwarz's Lemma (cont.)

Similar to problem 8 on 2010 May Prelim
Similar to problem 8 on 2006 May Prelim

7. Problems about Normality

Similar to problem 3 on Exam II In-Class
Similar to problem 9 on Exam II Take-Home
Similar to problem 2 on 2003 May Prelim
Similar to problem 8 on 2001 Prelim

8. Problems about Harmonic Functions

Similar to problem 6-7 on Exam II Take-Home
Similar to problem 4 on 1998 Final Exam
Similar to problem 8 on 2003 May Prelim

9. Problems about Weierstrauss Factorization Theorem

Similar to problem 5 on 1998 Final Exam
Similar to problem 8 on 2013 August Prelim
Similar to problem 5 on 2011 August Prelim
Similar to problem 10 on 2010 August Prelim
Similar to problem 10 on 2006 August Prelim

10. Problems about Convergence in the Topology of $A(G)$

Similar to problem 6 on 1998 Final Exam
Similar to problem 8 on 2013 August Prelim
Similar to problem 6 on 2012 August Prelim
Similar to problem 5 on 2011 May Prelim

11. Problems about Hurwitz's Theorem

Similar to problem 6 on 1998 Final Exam
Similar to problem 7 on 1998 Final Exam
Similar to problem 6 on 2012 May Prelim

12. Problems about the Domain of Analyticity of Infinite Series or Infinite Products

Similar to problem 8 on 1998 Final Exam
Similar to problem 8 on 2013 August Prelim
Similar to problem 8 on 2009 May Prelim
Similar to problem 3 on 2007 May Prelim
Similar to problem 6 on 2004 August Prelim